

CS4227_CS5722 - Software Architecture

Designing the architecture and Evaluating a Component-Based Research Management System

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Assignment 1

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Summary

Research Management Systems (RMS) are software platforms whose main objective is to provide various services for the management of research projects, data, resources, funding, and collaboration across academic, scientific, and industrial research environments. These systems are primarily used by Think Tank Research Centers, which are organizations that conduct research and advocacy on specific topics such as political, social, economic, scientific, or technological issues.

Given the importance of RMS, this document proposes software architecture for an RMS tailored for a Think Tank Research Center. Specifically, it explains the stakeholders and their needs, the software requirements, and the research and analysis of system requirements. A comprehensive explanation of the feasibility of this proposal is also given. The document ends with a table showing the contributions of each member who has worked on the project.

Stakeholder Needs

1. Researchers

- **Project Management:** Researchers need visual monitoring of ongoing projects, timelines, and milestones. This will encourage not only progress tracking but also effective collaboration, as researchers can view others' work.
- **Data Storage and Access:** Researchers need online and secure repositories where research data, findings, datasets, and literature reviews can be accessed. Additionally, the RMS should provide security features so researchers can manage the access levels of their work (e.g. private or public).
- **Collaboration Tools:** Researchers typically work in teams, which means that they need shared spaces for co-authoring papers, exchanging ideas with their peers, and discussing methodologies. In other words, the RMS should include tools that encourage and facilitate communication, document sharing, and real-time updates.

- **Resource Management:** Since researchers need access to resources such as funding, equipment, and personnel, it is important to provide features that allow them to track and allocate these resources in the shortest possible time.
- **Reporting Documentation:** Researchers need to generate reports and track their publications, so they can ensure not only proper citation but also documentation of their work. Furthermore, the RMS should incorporate private insights that allow researchers to check the popularity of their publications.

2. Analysts

- **Data Analysis Tools:** Analysts require easy-to-use tools that provide dataset processing, statistical analysis and modelling. Besides, RMS should store past reports and results that can be searched to allow analysts to identify insights through historical research data.
- **Reporting and Visualization:** Since analysts often deal with large datasets whose data is not normally easy to interpret and analyze, the RMS should have features for the creation of charts, graphs, and reports that help interpret data. This will not only improve the efficiency of these stakeholders, but also the public dissemination of their analysis.
- **Collaboration Tools:** Analysts often work with both researchers and policymakers to interpret data and identify trends. Therefore, it is important to incorporate tools that allow communication and collaboration between stakeholders.
- **Monitoring and Evaluation:** Analysts need to track the success of past policies or interventions to evaluate their effectiveness. Thus, the RMS should provide mechanisms that allow them to monitor and evaluate data.

3. Policymakers

- **Data Insights and Analysis:** Policymakers rely on research findings to guide their decisions. The RMS should offer visualizations and summaries of complex data, research outcomes, and policy implications.

- **Access to Research Findings:** Policymakers need fast, easy access to relevant reports, policy papers, and research recommendations. Therefore, the RMS should have mechanisms that allow quick searching and access of documents.
- **Collaboration Tools:** Similarly to the previous stakeholders, policymakers also need tools that encourage communication and collaboration between stakeholders.
- **Impact Tracking:** Since policymakers need to know how research influences the current policy decisions, it is important to offer tools to track the impact of published reports and analyze whether research has considered this policy.

Requirements Specification

The requirements for the Research Management System were gathered and prioritized using Jira, a "project management tool that helps you plan and track work across every team" (Atlassian, 2025). For further details on the Jira project, please refer to https://losedavidpb.atlassian.net/jira/software/projects/RMS/boards/3?atlOrigin=eyJpIjoiZjJiMzc5Mzc4OGFiNDBjY2I2M2EzYTQ0MjAxNzc0YzAiLCJwIjoiaiJ9. Additionally, this document also outlines these requirements using the Software Requirements Specification format (GeeksforGeeks, 2023), as follows:

Functional Requirements

1) User Registration

As a user, I want to be able to register for an account so that I can access the application's features.

2) User Login

As a registered user, I want to log in to my account using my email and password, so that I can access my personalized content and manage my account settings.

3) User Role

As an administrator, I want to assign roles to users, so that I can control access to different features and permissions based on the user's role (researchers, policymakers, data analysts, and administrators).

- **The system must support 4 roles:**
    - Administrator: Full access to all system features, user management, and data settings.
    - Researcher: Can access and edit research-related data, view reports, and submit new findings.
    - Policymaker: Can access and view reports, research findings, and insights but cannot modify research data.
    - Data Analyst: Can analyze data, create reports, and generate data visualizations, but cannot modify core research findings.

4) Notifications

As a user, I want to receive notifications for relevant activities (new discussions, file updates, etc.), so that I stay informed and can take action as needed.

- Users receive notifications for new discussion threads.
- Users receive notifications when a document they're collaborating on is updated.
- Users can configure notification preferences (e.g., frequency or type of notifications).

5) Tagging Features

As a user, I want to tag users in discussion threads or documents, so that I can directly notify them and bring attention to specific items.

- Users can tag others in discussions and document comments using the "@" symbol.

- Tagged users receive notifications about the mention.
- Tags are clickable and lead to the profile of the tagged user.

6) Discussion Boards

As a user, I want to be able to initiate and participate in topic-specific discussions, so that I can collaborate with others, share knowledge, and brainstorm.

- Users can create discussion threads on specific topics.
- Users can reply to discussion threads.
- Users can view all discussion threads they are part of.
- The system notifies users of new replies to threads they are following.

7) Real-Time Document Collaboration

As a user, I want to collaborate on documents in real-time with others, so that we can work together efficiently and update files simultaneously.

- Users can edit documents simultaneously.
- Changes made by any user are immediately reflected for all users in real-time.
- The system tracks changes (e.g., via a version history or change log).

8) Document Version Control

As a user, I want to have version control for documents, so that I can track changes, revert to previous versions, and avoid conflicts.

- Each document version is automatically saved.

- Users can view and revert to previous versions of a document.
- Users can see who made specific changes in version history.

9) Research Project Tracking

As a researcher, I want to submit research proposals with relevant details and track their progress through defined stages so that I can ensure my proposals are reviewed, approved, and completed on time.

- **Proposal Submission**:
    - Researchers can submit research proposals by providing the required details (e.g., title, description, budget, timeline, etc.).
    - The system should validate the completeness of the proposal and ensure all necessary information is included before submission.
- **Proposal Review and Approval**:
    - Administrators can view submitted proposals and review the provided details.
    - Administrators can approve or reject proposals, and if rejected, provide feedback for revisions.
    - Researchers are notified of the approval or rejection status.
- **Progress Tracking**:
    - Each proposal has a progress status that can be tracked across different stages: "Under Review," "In Progress," and "Completed."
    - Researchers can update the progress of their projects within the defined stages, moving from "Under Review" to "In Progress," and finally to "Completed."
    - Administrators can view the current progress of all research proposals and ensure milestones are met.

- **Notifications**:
    - Researchers receive notifications when their proposal status changes (e.g., when it's approved, rejected, or moved to another stage).
    - Administrators receive notifications of new submissions awaiting review.

10) Required Integrations

The application must be integrated with external databases such as Google Scholar, Institutional Repositories, Scopus, and ORCID to access research papers.

- The application must allow users to search for and retrieve research papers from Google Scholar, Scopus, and Institutional Repositories through their respective APIs or data sources.
- The system must support authentication and authorization mechanisms to interact with ORCID for retrieving author profiles and related publications.
- The application must display relevant metadata (title, authors, publication date, etc.) from research papers retrieved from these external sources.
- The system should handle errors gracefully if external systems are unavailable, providing informative error messages to users.

11) Reporting Analytics

As a researcher, I want to generate progress reports, track citations, view dashboards, and receive AI-based suggestions, so that I can easily monitor my research progress, manage references, and get relevant recommendations for improving my work.

- **Progress Reports**:
    - The system must generate progress reports showing the status of ongoing research projects, including key milestones and achievements.

    - Users can filter progress reports by date, project, or status (e.g., "In Progress," "Completed").

- **Citation Tracking**:
    - The system must track citations for each research paper, pulling citation data from external databases like Google Scholar or Scopus.
    - Users can view citation counts and trends over time for their published research.
- **Dashboards**:
    - The application must provide a customizable dashboard displaying key metrics like project progress, citation counts, upcoming deadlines, and AI recommendations.
    - Users should be able to add or remove widgets from the dashboard according to their preferences.
- **AI-Based Suggestions**:
    - The system must provide AI-driven suggestions for relevant literature, research methods, or collaborators based on the user's research activity and past work.
    - Users can accept, reject, or dismiss these suggestions, and the system should refine future suggestions based on feedback.

Non-functional Requirements

1) Security Requirements

Given the sensitive nature of research data, **security** is a critical aspect of the RMS. The system must ensure:

- **Authentication & Authorization:** All users must authenticate using **multi-factor authentication (MFA)** where applicable. Role-based access control (RBAC) should restrict access to data based on user roles (Researcher, Policymaker, Analyst, and Administrator).

- **Data Encryption:** All sensitive data must be stored using **AES-256 encryption** and transmitted using **TLS 1.3**.
- **Compliance with GDPR & Data Protection Standards:** User data must be stored securely, ensuring compliance with **GDPR** and other relevant **data protection regulations**.
- **Audit Logging:** The system should maintain **logs** of all critical actions (e.g., document updates, data exports, role modifications) with timestamps and user IDs.
- **Backup & Disaster Recovery:** The system must perform **daily backups** and support recovery within **2 hours** in case of failures.
- **Secure API Integrations:** External integrations (Google Scholar, ORCID, Scopus) must follow **OAuth 2.0** authentication and ensure API keys are securely managed.

2) Software Quality Attributes

To ensure high quality and maintainability, the RMS must adhere to the following **software quality attributes**:

- **Scalability:** The system should support the growing number of users and datasets by ensuring a modular and distributed architecture.
- **Usability:** The UI must be **intuitive, responsive, and accessible** (meeting **WCAG 2.1 AA** accessibility standards).
- **Maintainability:** The system architecture should be **modular** to allow for easy updates and feature enhancements. Code should follow **SOLID** principles to facilitate maintainability.

3) Business Rules

- **User Access Restrictions:** Only authorized personnel can access sensitive research data.
- **Publication Approval Workflow:** All research reports must be **reviewed and approved** before publication.

- **Data Retention Policy:** Research data should be stored for **at least 5 years** and then archived or deleted as per policy.
- **Collaboration Rules:** Researchers can share documents **only within designated teams or external collaborators** with appropriate permissions.
- **System Updates:** The RMS should receive **quarterly updates** for security patches and feature enhancements.
- **Funding & Budget Allocation:** Researchers must submit funding requests through the RMS, subject to **admin approval**.

Research & Analysis of System Needs

Research management Systems (RMS) are software applications that assist universities, research institutes and Think Tanks Research centres with project management, funding management and data sharing tools. This section explores some of the most popular RMS platforms that were analysed for the requirements gathering. For each of these RMS, their strengths and weaknesses are listed.

1. Elsevier Pure

Elsevier Pure is among the most complete RMS platforms and is used by universities and big research institutions. It is the hub for all research data, funding and publications (Elsevier, 2025).

- **All-in-One Research Management:** Pure tracks everything from grant applications to publication records and collaborations for researchers.
- **Data Analytics and Reporting:** Integrates with Elsevier's Scopus & SciVal to easily visualize research impact and citation patterns.
- **Collaboration Tools:** Researchers can collaborate inside the system and monitor improvement and communicate updates in real time.
- **Open Access Tracking - Compliance:** Assists institutions to ensure that research outputs meet open access policies and funding requirements.

Strengths:

- Strong statistics and reporting software.
- Easy integration with Elsevier databases.
- Institutional dashboards may be customized.

Weaknesses

- Expensive, which makes it unavailable for small businesses.
- Could be complicated to put together and calls for technical knowledge.

2. Symplectic Elements

Symplectic Elements is a tool for universities to monitor research outputs, impact and funding. It is especially popular in universities which require institutional repositories (Symplectic, 2025).

- **Automatic Publication Tracking:** Elements aggregates information from external sources including Scopus and PubMed to keep researcher profiles up to date.
- **Grants Management - Funding:** Designed to help researchers monitor applications and compliance requirements.
- **ORCID Integration:** Connects with ORCID to synchronize researcher profiles and publication records.
- **Support for Institutional Repository:** Integrates with platforms like EPrints and DSpace for data storage.

Strengths:

- Automates information gathering by outside sources.
- Strong assistance for open access & compliance monitoring.
- Customizable for institution requirements.

Weaknesses:

- Requires personalization & administration control.
- The interface may be intimidating for new users.

3. Converis:

Converis is a web-based research information system supporting everything from grant applications to publication tracking. It is utilized by universities and government research organizations (Clarivate, 2025).

- **End-to-End Lifecycle Management of Scientific Research:** Tracks research projects from conception to publication.
- **Comprehensive Grant Management:** Designed to help researchers and administrators with managing funding applications and approvals.
- **Bibliometric Analysis:** Connects with Web of Science and Scopus to offer insights into research impact.
- **Workflows That Can Be Easily Customized:** Institutions can tailor Converis to fit their specific research management procedures.

Strengths:

- Highly customizable & caters to different research requirements.
- Powerful analytics to measure research impact.
- Facilitates collaboration among various investigation areas.

Weaknesses:

- Expensive and resource-intensive to implement.
- Requires specific technical help for efficient usage.

4. Figshare

Figshare is a public access research information management system enabling sharing and discovery of research outputs. It's used by universities and individual researchers desiring to make work public (Figshare, 2025).

- **Sharing & Data Storage:** Permanent DOI links enable scientists to store datasets, figures, and publications.
- **Version Control & Metadata Management:** Ensures documentation of research outputs which can be easily updated over time.
- **Institutional Repositories:** Universities also can develop branded repositories for their research data.
- **Support for Open Science:** Figshare is built for open access publishing to help institutions meet funder demands.

Strengths:

- Encourages open science & sharing of knowledge.
- User friendly interface for researchers.
- Supports integration of various other research processes.

Weaknesses:

- Limited in features besides storage.
- Storage restrictions because of institutional licensing.

5. Open Science Framework (OSF)

OSF is an open-source and free research management program enabling collaboration and openness in research. It is utilized by individual researchers and small teams (OpenScience, 2025).

- **Project Management:** Organizes work, assigns tasks, and monitors progress.
- **Collaboration Tools:** Allows remote sharing of documents and version control.
- **Preprint Hosting:** Allows researchers to share early results before publication.
- **Third-Party Integrations:** Works with services like GitHub, Dropbox and Google Drive.

Strengths:

- Open source (Free for researchers).
- Strong focus on reproducibility and clear homework.
- Encourages interdisciplinary cooperation.

Weaknesses:

- Lacks innovative fund & grant administration tools.
- Requires institutional support for large scale adoption.

Feasibility

Implementing a research management system using Java Spring Boot and React is highly feasible for our requirements. Java Spring Boot provides a robust and scalable backend framework that supports microservices, RESTful API creation, and seamless integration with various databases and third-party services. On the front end, React offers dynamic, component-based architecture that enables the creation of highly responsive user interfaces. Together, these technologies facilitate modular, maintainable, and performant systems that can be adapted to meet evolving stakeholder requirements. With proper planning and iterative development practices, the implementation can be accomplished efficiently within typical budget and timeline constraints, making it a practical choice for our research management solution.

Team contributions

Group members	Week 1	Week 2	Week 3
Shane Barden - 24033944	Literature review	Identify stakeholders	Requirements Specification
Ivor D Souza - 24177431	Literature review	Functional requirements gathering	Feasibility, functional, non-functional requirements

Manjeshwar Aniruddh Mallya – 24124133	Literature review	Non-functional requirements gathering	Research and Analysis of System Needs
David Parreño Barbuzano - 24196223	Literature review	Functional requirements gathering	Summary, Stakeholders Needs
Rohan Sikder - 24165816	Literature review	Non-functional requirements gathering	Research and Analysis of System Needs

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